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RTCA/DO-160G

Environmental Conditions and Test Procedures for Airborne Equipment

Section 5

Temperature Variation

Important Notice

Information contained in these sections is pertinent to all test procedures described in the other sections of this document. Further, Appendix A is applicable for identifying environmental tests performed.

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5.0 Temperature Variation

5.1 Purpose of the Test

This test determines performance characteristics of the equipment during temperature variations between high and low operating temperature extremes. The applicable categories for flight operations are shown in Table 4-1. For categories A, B, and C, it is required that the equipment be subjected to the temperature variation test in 5.3.1 when such equipment is tested according to the procedures contained in Paragraphs 4.5.1, 4.5.2, 4.5.3, and 4.5.4. For categories S1 and S2, it is required that the equipment be subjected to the temperature shock test per Paragraph 5.3.2 or 5.3.3.

Note: The test is not intended to verify the behavior of the equipment in wet or icing conditions. In conducting this test, the test chamber may incorporate the capability of controlling or altering humidity to the extent that condensation is minimized or does not occur.

5.2 Temperature Change Rates

The rates applicable to the temperature variation procedures defined in Subsection 5.3 are as follows:

Category A - For equipment external to the aircraft or internal to the aircraft:

10 degrees Celsius minimum per minute.

Category B - For equipment in a non-temperature-controlled or partially temperature controlled internal section of the aircraft:

5 degrees Celsius minimum per minute.

Category C - For equipment in a temperature-controlled internal section of the aircraft:

2 degrees Celsius minimum per minute.

Category S1 - For equipment external to the aircraft or internal to the aircraft:

Known rate of change greater than 10 degrees Celsius per minute. The rate of change shall be noted in the Qualification Form.

Category S2 - For equipment external to the aircraft or internal to the aircraft:

Unknown rate of change greater than 10 degrees Celsius per minute.

Note: *1) Equipment qualified to Category B is considered to have met Category C.*

2) Rates of change greater than 10 degrees Celsius minimum per minute are considered as temperature shocks. Categories S1 and S2 are intended to evaluate the effects of more rapid or sudden temperature changes to the

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equipment. The outer surfaces, regions, or interfaces of the equipment may be of primary interest during temperature shock as the internal regions can have comparatively long time constants to reach stabilization.

3) For Categories S1 and S2, Thermal Shock, Category S2 shall be used unless the temperature change rate is known, then Category S1 may be used.

5.3 Test Procedures

5.3.1 Test Procedure Categories A, B, and C

The temperature variation test (except for Categories S1 and S2) can be combined to include the procedures of the ground survival low temperature test and short-time operating low temperature test, Paragraph 4.5.1, the operating low temperature test, Paragraph 4.5.2, the ground survival high temperature test short-time operating high temperature test, Paragraph 4.5.3, and the operating high temperature test, Paragraph 4.5.4. The following procedures shall apply:

a. If the test is a combined test, proceed in accordance with Paragraph 4.5.1, which describes the ground survival low temperature test and the short-time operating low temperature test, and Paragraph 4.5.2, the operating low temperature test without transition to ambient temperature. After completion of the test defined in Paragraph 4.5.1 and 4.5.2, proceed to Subparagraph c. If the test is not a combined test, commencing at ambient temperature with the equipment operating, lower the temperature in the chamber towards the operating low temperature level at the applicable rates specified in Subsection 5.2.

b. Stabilize the equipment in the operating mode at this operating low temperature level.

c. Raise the temperature in the chamber towards the operating high temperature at the applicable rate specified in Subsection 5.2. During this temperature change, DETERMINE COMPLIANCE WITH THE APPLICABLE EQUIPMENT PERFORMANCE STANDARDS.

d. Stabilize the equipment at the operating high temperature. If this is a combined test, proceed in accordance with Paragraph 4.5.3, the ground survival high temperature test and short-time operating high temperature test, and subsequently Paragraph 4.5.4, the operating high temperature test without transition to ambient temperature. Maintain the equipment in a non-operating state for a minimum of 2 minutes.

e. Turn the equipment on and lower the temperature in the chamber towards the operating low temperature level at the applicable rate specified in Subsection 5.2. During this temperature change DETERMINE COMPLIANCE WITH THE APPLICABLE EQUIPMENT PERFORMANCE STANDARDS.

f. Stabilize the equipment temperature with the chamber at the operating low temperature, and then operate the equipment for at least one hour. Then turn off the equipment for 30 minutes, and restart the equipment while maintaining the chamber at the operating low temperature.

g. Change the temperature of the chamber towards the ambient temperature at the applicable rate specified in Subsection 5.2.

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h. Stabilize the chamber and the equipment at ambient temperature. DETERMINE COMPLIANCE WITH THE APPLICABLE EQUIPMENT PERFORMANCE STANDARDS in the second, or last, cycle as appropriate.

A minimum of two cycles (a. through h. above) shall be accomplished. If complete determination of compliance with applicable equipment performance standards can be accomplished during each temperature change period of a single cycle, then testing is required during the second cycle only. If the time during a temperature change period does not allow for complete determination of compliance with applicable equipment performance standards, a sufficient number of cycles shall be accomplished so that complete compliance can be determined. When temperature rise induces a potential risk of condensation on the equipment under test, the humidity level of the air in the chamber should be controlled to eliminate this condensation. The test profile is shown graphically in Figure 5-1.

It is permissible to allow additional time with the equipment un-powered at ambient conditions, between the first and second, and subsequent, cycles to accommodate test facility and personnel schedules. Resumption of the second, or subsequent, cycle must be accomplished within 24 hours. No other environmental testing may be performed between cycles.

Note: If this is a combined test, it is not necessary to repeat the Ground Survival Low Temperature, Short-Time Operating Low Temperature, Ground Survival High Temperature, and Short Time Operating High Temperature tests as defined in steps a. and d. above during the second cycle.

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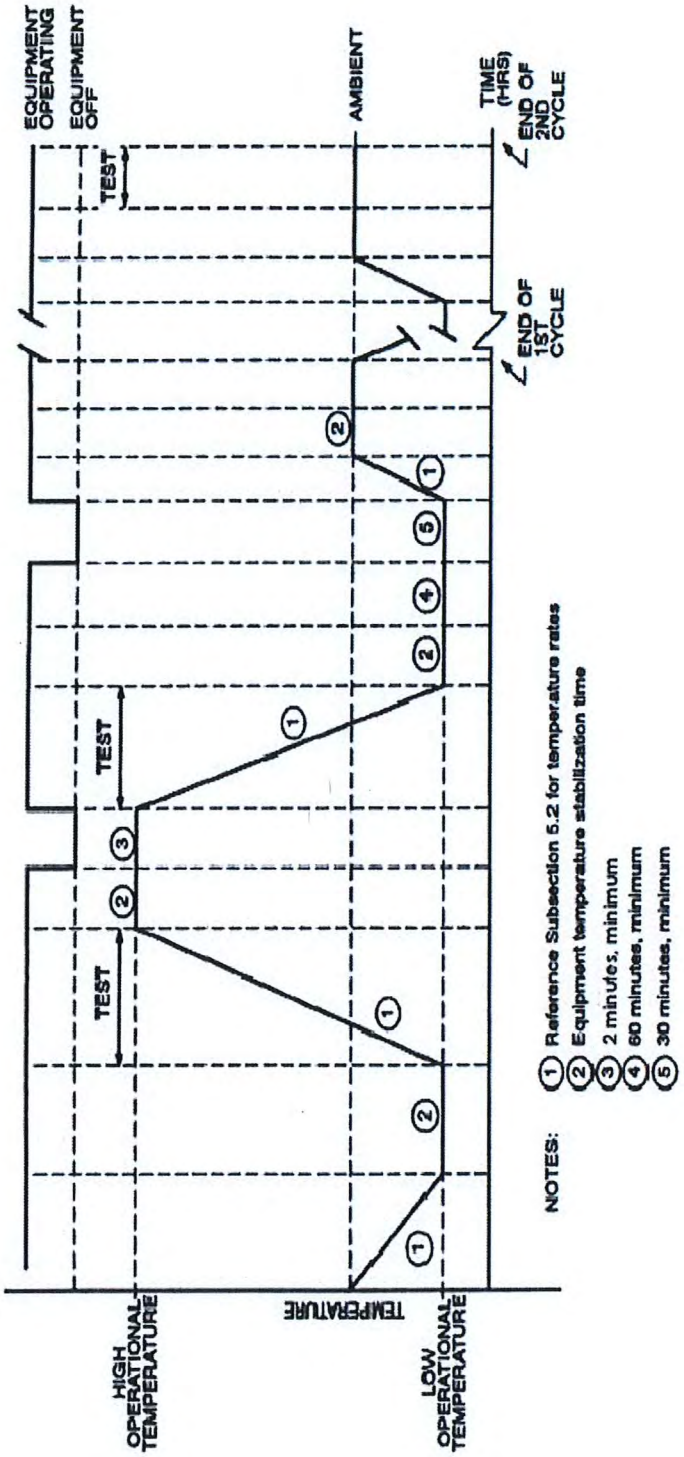

FIGURE 5-1 TEMPERATURE VARIATION TEST

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5.3.2 Test Procedure Category S1

The temperature variation test for Category S1 should not be combined with other temperature testing.

The following procedures shall apply:

a. Commencing at ambient temperature with the equipment operating, lower the temperature in the chamber towards the operating low temperature level at the applicable rates specified in Subsection 5.2.

b. Stabilize the equipment, if workable, at this operating low temperature level.

c. Raise the temperature in the chamber towards the operating high temperature at the applicable rates specified in Subsection 5.2. During this temperature change, DETERMINE COMPLIANCE WITH THE APPLICABLE EQUIPMENT PERFORMANCE STANDARDS.

d. Stabilize the equipment at the operating high temperature, if workable. Maintain the equipment in a non-operating state for a minimum of 2 minutes.

e. Turn the equipment on and lower the temperature in the chamber towards the operating low temperature level at the applicable rates specified in Subsection 5.2. During this temperature change DETERMINE COMPLIANCE WITH THE APPLICABLE EQUIPMENT PERFORMANCE STANDARDS.

f. Stabilize the equipment temperature, if workable, with the chamber at the operating low temperature, and continue to operate the equipment for at least one hour. Then turn off the equipment for 30 minutes, and restart the equipment while maintaining the chamber at the operating low temperature.

g. Change the temperature of the chamber towards the ambient temperature at the applicable rates specified in Subsection 5.2.

h. Stabilize the chamber and the equipment at ambient temperature, if workable. DETERMINE COMPLIANCE WITH THE APPLICABLE EQUIPMENT PERFORMANCE STANDARDS in the second, or last, cycle as appropriate.

A minimum of two cycles (a. through h. above) shall be accomplished. If complete determination of compliance with applicable equipment performance standards can be accomplished during each temperature change period of a single cycle, then testing is required during the second cycle only. If the time during a temperature change period does not allow for complete determination of compliance with applicable equipment performance standards, a sufficient number of cycles shall be accomplished so that complete compliance can be determined. When temperature rise produces condensation on the equipment tested, the humidity level of the air in the chamber should be controlled to eliminate this condensation.

5.3.3 Test Procedure Category S2

The temperature variation test for Category S2 shall not be combined with other temperature testing.

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